Plasticity, nutrition, and breeding

We want to understand how nutrition affects breeding date, which can be written as follows:

$$y = a + bn + e. \tag{1}$$

Specifically, we can try to mechanistically model optimal breeding date as a function of nutrition by representing fitness as a product of fecundity and offspring survival. First, we assume that birds with higher access to resources have increased offspring survival at an earlier date, where the offspring survival function S can be written as:

$$S(n, y) = \exp\left(\alpha_S n - \frac{(\theta_S - y)^2}{2\omega_y^2}\right), \tag{2}$$

where y represents the breeding date; α_S represents the effect of nutritional condition on offspring survival; n represents nutritional condition; $\theta_S = A_S + B_S n$ represents the optimal breeding date based on offspring survival; and ω_y represents the width of survival function Second, we assume that birds with higher access to resources increased fecundity, where the fecundity function F can be written as:

$$F(n, y) = \exp\left(\alpha_F n + cy\right), \tag{3}$$

where α_F represents the effect of nutritional condition on fecundity and c represents a scaling factor. To keep the computation simple for now, we're not going to include the cost of plasticity. Then, the overall fitness can be represented as a product of these three terms:

$$W(n, y) = \exp\left(\alpha n - \frac{(\theta_S - y)^2}{2\omega_y^2} + cy\right), \tag{4}$$

where $\alpha = \alpha_S + \alpha_F$. Here, the major difference from Hadfield and Reed's result is the cy term, which accounts for increase in fitness through increase in fecundity. Therefore, the actual breeding date must balance the early breeding effect from offspring survival with later breeding effect from fecundity.

Now, we can take the average across breeding date y:

$$\bar{W}(n, \bar{y}) \tag{5}$$

$$= \int p(y) W(n, y) \,\mathrm{d}y \tag{6}$$

$$= \exp\left(\alpha n\right) \int p(y) \exp\left(-\frac{(\theta_S - y)^2}{2\omega_y^2} + cy\right) \mathrm{d}y \tag{7}$$

$$= \exp\left(\alpha n\right) \frac{1}{\sqrt{2\pi\sigma^2}} \int \exp\left(-\frac{(y - \bar{a} - \bar{b}n)^2}{2\sigma^2}\right) \exp\left(-\frac{(\theta_S - y)^2}{2\omega_y^2}\right) \exp\left(cy\right) \,\mathrm{d}y \tag{8}$$

$$= \exp\left(\alpha n\right) \frac{1}{\sqrt{\sigma_i^2}} \times \tag{9}$$

$$\sqrt{\frac{\sigma^2 \omega_y^2}{\sigma^2 + \omega_y^2}} \exp\left(-\frac{-c^2\sigma^2\omega_y^2 - 2\theta_S(c\sigma^2 + \bar{a} + \bar{b}n) + 2c(\bar{a} + \bar{b}n)\omega_y^2 + \theta_S^2 + (\bar{a} + \bar{b}n)^2}{2(\sigma^2 + \omega_y^2)}\right) \tag{10}$$

$$= \exp\left(\alpha n\right) \sqrt{\omega^2 \gamma} \times \tag{11}$$

$$\sqrt{\frac{\sigma^2 \omega_y^2}{\sigma^2 + \omega_y^2}} \exp\left(-\frac{-c^2\sigma^2\omega_y^2 - 2\theta_S(c\sigma^2 + \bar{a} + \bar{b}n) + 2c(\bar{a} + \bar{b}n)\omega_y^2 + \theta_S^2 + (\bar{a} + \bar{b}n)^2}{2(\sigma^2 + \omega_y^2)}\right) \tag{12}$$

$$\tag{13}$$

, where $\sigma^2 = G_{aa} + 2G_{ab}n + G_{bb}n^2 + \sigma_e^2$ and $\gamma = 1/(\sigma^2 + \omega_y^2)$

We can also model the cost of plasticity:

$$C(n, y, b) = \exp\left(-\frac{b^2}{2\omega_b^2}\right), \tag{14}$$

where b represents plasticity and ω_b represents the width of cost of plasticity. Then, the overall fitness can be represented as a product of these three terms:

$$W(n, y, b) = \exp\left(\alpha n - \frac{(\theta_S - y)^2}{2\omega_y^2} + cy - \frac{b^2}{2\omega_b^2}\right), \tag{15}$$

where $\alpha = \alpha_S + \alpha_F$. This time, we can take the average across breeding date y and plasticity b:

$$\bar{W}(n, \bar{y}, \bar{b}) \tag{16}$$

$$= \iint p(y, b) W(n, y, b) \, \mathrm{d}y \, \mathrm{d}b \tag{17}$$

$$= \exp\left(\alpha n\right) \iint p(y, b) \exp\left(-\frac{(\theta_S - y)^2}{2\omega_y^2} + cy - \frac{b^2}{2\omega_b^2}\right) \, \mathrm{d}y \, \mathrm{d}b \tag{18}$$

$$= \exp\left(\alpha n\right) \int p(b) \exp\left(-\frac{b^2}{2\omega_b^2}\right) \int p(y|b) \exp\left(-\frac{(\theta_S - y)^2}{2\omega_y^2} + cy\right) \, \mathrm{d}y \, \mathrm{d}b \tag{19}$$

$$= \exp\left(\alpha n\right) \int p(b) \exp\left(-\frac{b^2}{2\omega_b^2}\right) \frac{1}{\sqrt{2\pi\sigma_i^2}} \times \tag{20}$$

$$\int \exp\left(-\frac{(y - \bar{a} - bn)^2}{2\sigma_i^2}\right) \exp\left(-\frac{(\theta_S - y)^2}{2\omega_y^2}\right) \exp\left(cy\right) \, \mathrm{d}y \, \mathrm{d}b \tag{21}$$

$$= \exp\left(\alpha n\right) \int p(b) \exp\left(-\frac{b^2}{2\omega_b^2}\right) \frac{1}{\sqrt{\sigma_i^2}} \times \tag{22}$$

$$\sqrt{\frac{\sigma_i^2 \omega_y^2}{\sigma_i^2 + \omega_y^2}} \exp\left(-\frac{-c^2\sigma_i^2\omega_y^2 - 2\theta_S(c\sigma_i^2 + \bar{a} + bn) + 2c\mu\omega_y^2 + \theta_S^2 + (\bar{a} + bn)^2}{2(\sigma_i^2 + \omega_y^2)}\right) \tag{23}$$

$$= \exp\left(\alpha n\right) \int p(b) \exp\left(-\frac{b^2}{2\omega_b^2}\right) \sqrt{\omega^2 \gamma} \frac{1}{\sqrt{\sigma_i^2}} \times \tag{24}$$

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